2. ACCRETION METHOD AND TESTS

For all our simulations, we use the adaptive mesh refinement code FLASH 4 with its directionally unsplit hydrodynamic solver and a fixed gravitational potential (Fryxell et al. 2000; Dubey et al. 2008). AGN are treated as particles in our simulations. For the runs in this paper we place one AGN particle at the center of the grid at rest to approximate an isolated cluster environment, but all the accretion and feedback algorithms are connected to properties of the AGN particle in the particle's rest frame and thus can be used in cosmological simulations with multiple AGN particles.

2.1. *Sink Particle Algorithm*

In most previous work involving sink particles, some form of reset of the gas properties within the sink particle radius is performed to conserve mass, energy and momentum. The reset usually involves an arbitrary kernel function and some correction parameters. Motivated by this work, we describe our model as follows.

2.1.1. *Measuring the Accretion Rate*

In our model, the accretion configuration allows us to capture the infall of low angular momentum gas blobs and molecular clouds that are predicted by chaotic cold accretion. Most of the CCA condensations occur around $10^4 r_{\rm B}$ ($r_{\rm B} \equiv GM/c_{s,\infty}^2$ is the Bondi radius, discussed further in 2.2.1) and start major collisions around $10-100 r_{\rm B}$, contributing to the AGN fueling process (Gaspari et al. 2017). Capturing this process is crucial to studying the self-regulation of AGN feedback. One of the major challenges is to distinguish the outflow from eddies generated by the infalling clouds, as well as from the disk wind and the jet.

The accretion rate $\dot{M}$ is calculated by estimating the mass flux across a spherical control surface centered on the AGN and having a radius $R_{\rm cs}$. This is more physically realistic than calculating the mass gain across a number of cells or just using the Cartesian flux returned from the Riemann solver. We use a Monte Carlo method, in which we choose N random points on the surface with different angles (θ_i, ϕ_i), and sum the mass flow over these points to obtain

$$\dot{M} = -R_{\rm cs}^2 \int d\Omega\, \rho \mathbf{v} \cdot \hat{\mathbf{n}} \approx -\frac{4\pi R_{\rm cs}^2}{N} \sum_{i=1}^{N} \rho_i v_{r,i} \ . \qquad (1)$$

Here ρ_i, $\mathbf{v}_i$, and $\mathbf{n}_i$ are the density, velocity, and unit outward normal vector for the ith control surface point. To ensure the points are uniformly distributed on the sphere, we use uniform deviates u, v and calculate (θ_i, ϕ_i) as $\theta_i = 2\pi$u and $\phi_i = \arccos$ (2v-1).

Only inflow points (where $v_{r,i} = \mathbf{v}_i \cdot \hat{\mathbf{n}}_i \leq 0$) are counted, since we need to include all the material that has been falling onto the SMBH but exclude the jet. Two methods are considered: one is to count zero at points where there is an outflow, so that there are fewer than N inflow points; another is to discard each outflow point and keep searching until we have N inflow points. For our current jet configuration, we have run both methods and find the difference to be negligible since the jet only covers a small fraction of the control surface. However, since in the general case outflow can take place over a larger portion of the control surface, we use the second method in the calculations described here to minimize noise in the estimate of the integral.

We tested our flux calculation by initializing a Bondi profile setup (details of which are outlined in § 2.2.1) on a uniform grid, and compared the results from the different flux methods to the analytical Bondi formula (Equation 17). We estimated the uncertainty in this accretion rate calculation by varying two parameters: the number of random points, and the interpolation method from the mesh onto the control surface. For the latter, we used either tri-linear or nearest-neighbor interpolation. Figure 1 shows the relative error with respect to Equation 17 as a function of control surface radius and number of samples $N = N_{\rm samp}$. Tri-linear interpolation gives smaller error for a given number of sample points and $R_{\rm cs}$ value. The error scales with $\sqrt{N}$ to a certain radius ($R_{\rm cs} \sim 250\Delta x$, the maximum box size used), and the variation of error decreases with increasing number of sample points. We adopt $N = 1000$ and the tri-linear interpolation method for later calculations. The same approach can also be used to compute momentum and angular momentum fluxes when they are included in the SMBH configuration.

2.1.2. *Removing Gas from the Grid*

In our simulations we are not directly resolving the accretion disk, so after the accretion rate is calculated we must remove the gas accreted and effectively create an inner boundary that absorbs the inflow onto the black hole. Analogously to common sink particle algorithms in AMR simulations, this is handled by resetting the gas density and pressure within a spherical region of radius $R_{\rm reset}$ around the black hole. The gas velocity is left unchanged, while the internal energy is reset using the equation of state.

To remove gas from the grid, we use a multiplicative resetting kernel whose value equals one at $R_{\rm reset}$ to make the modified values continuous across the surface of the resetting region. The resetting kernel is defined to be

$$\psi(r) = \psi_0 \left[\omega(r) - e^{-1}\right] + 1 \ , \qquad (2)$$

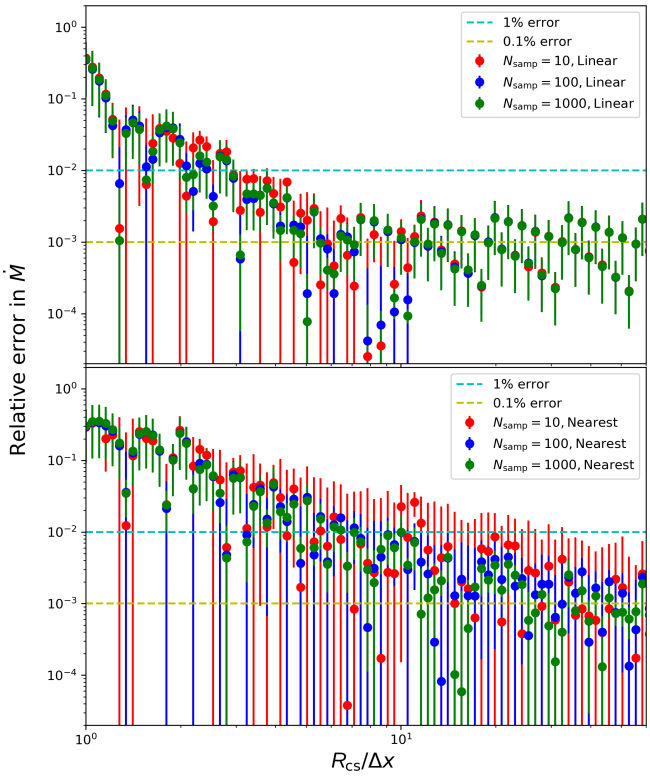

Figure 1. Error in the measured accretion rate (with respect to the analytical Bondi formula) and its relationship with the control surface radius in our accretion calculation, with different number of samples, and different in-grid interpolation techniques. The dashed lines represent 1% and 0.1% error respectively.

where

$$\omega(r) = \exp\left(-r^2/R_{\rm reset}^2\right) \ . \tag{3}$$

Here ψ_0 is the normalization factor, which is calculated by requiring that the total gas removed be equal to the amount of gas accreted in a timestep, $\dot{M}\Delta t$. We set the new density in zone ijk to be (with n representing the current timestep and $n+1$ the next step)

$$\rho_{ijk}^{\rm n+1} = \rho_{ijk}^n \psi(r_{ijk}) \ , \tag{4}$$

where r_{ijk} is the radial position of the zone center. We require that the mass removed satisfy

$$-\dot{M}\Delta t = \int \psi \rho^n dV - \int \rho^n dV \ . \tag{5}$$

Given the form of ψ, we have for the normalization factor

$$\psi_0 = \frac{\dot{M}\Delta t}{e^{-1}\int \rho^n\, dV - \int \omega \rho^n\, dV} \ , \tag{6}$$

so that

$$\rho^{n+1} = \rho^n \left(1 + \frac{\dot{M}\Delta t\left(\omega - e^{-1}\right)}{e^{-1}\int \rho^n\, dV - \int \omega\rho^n\, dV}\right) \ . \tag{7}$$

The volume integrals in these expressions are computed by summing the respective quantities within the cells inside the spherical reset region.

To ensure that no zones within the resetting region fall below a minimum density $\rho_{\rm min}$, we impose an additional constraint on the resetting kernel. For each zone inside the reset radius, we require

$$\rho_{ijk}^{n+1} = \rho_{ijk}^n[1 - \psi_0(\omega_{ijk} - e^{-1})] \geq \rho_{\rm min} \ . \tag{8}$$

Each of these zones thus has an allowed maximum value for the normalization factor ψ_0:

$$\psi_{0,{\rm max},ijk} = \frac{1 - \rho_{\rm min}/\rho_{ijk}^n}{\omega_{ijk} - e^{-1}} \ . \tag{9}$$

Hence our final choice for ψ_0 is

$$\psi_{0,{\rm final}} = \min\left\{\psi_0, \min_{r_{ijk}\leq R_{\rm reset}} \psi_{0,{\rm max},ijk}\right\} \ . \tag{10}$$

In our simulations, we have set $\rho_{\rm min}$ to be 10^{-33} g cm^{-3} to avoid numerical instability.

Similarly, we can reset the pressure according to the change of the internal energy density within the control surface per time step. We note however, since the transport equation for pressure has both advection and compression/expansion parts, simply measuring a flux for the pressure is not a correct approach. However, we can still use the change in pressure within the control region due to the hydrodynamic solver to determine how much internal energy is accreted. Assume that the change in energy density satisfies

$$-\dot{u}\Delta t = -\frac{1}{\gamma - 1}\int (P^* - P^n)\, dV \ , \tag{11}$$

where P^* is the updated pressure returned from the hydrodynamics equations and the physical processes that follow (e.g. radiative cooling), and γ is the adiabatic index. Here we have adopted $\gamma = 5/3$ for non-relativistic gas. We thus have

$$P^{n+1} = P^*\left(1 + \frac{(\gamma-1)\dot{u}\Delta t\left(\omega - e^{-1}\right)}{e^{-1}\int P^*\, dV - \int \omega P^*\, dV}\right) \tag{12}$$

This will also be adapted in case we need to modify ψ_0 in order not to over-remove the mass accreted.

At the end of the timestep, we calculate the integrals by summarizing the gas property values in the region within $R_{\rm reset}$, then calculate the fluxes and reset the gas properties. Then we calculate the new integral values that will be used to evaluate the fluxes returned from the physical processes in the next timestep. We leave the velocity components unchanged to avoid any numerical instabilities. This creates a non-reflecting boundary, and the gas properties inside the boundary do not impact the structure of the flow outside.

2.2. *Tests of the Accretion Model*

As verification tests, we run two sets of controlled experiments: the Bondi problem and the Bondi-Hoyle-Lyttleton (BHL) problem. The setup for each problem allows us to compare the simulation outputs to analytical results and published simulation work.

2.2.1. *Bondi Test*

For this test, we assume that the accretion is spherically symmetric, and that a point mass M is embedded in the ambient gas. This problem was first addressed in Bondi (1952). Far from the accreting object, the medium has a uniform density ρ_∞ and a uniform pressure P_∞. For adiabatic gas where $P \propto \rho^\gamma$, the sound speed far away should have the value $c_{s,\infty} = \sqrt{\gamma P_\infty/\rho_\infty}$. For isothermal gas, instead we have $c_{s,\infty} = \sqrt{P_\infty/\rho_\infty}$.

We define the dimensionless variables

$$x \equiv \frac{r}{r_{\rm B}},\ u \equiv \frac{|v|}{c_{s,\infty}},$$
$$\alpha \equiv \frac{\rho}{\rho_\infty},\ \lambda \equiv \frac{\dot{M}}{4\pi\rho_\infty (GM)^2/c_{s,\infty}^3}, \tag{13}$$

where λ is the dimensionless accretion rate, and $r_{\rm B}$ is the Bondi radius $r_{\rm B} = GM/c_{s,\infty}^2$. Then we can solve the mass and momentum conservation equations in a dimensionless form

$$u\alpha x^2 = \lambda \tag{14}$$

$$\frac{u^2}{2} + H(\alpha) - \frac{1}{x} = 0 \tag{15}$$

where $H(\alpha)$ is given in polytropic flow by

$$H(\alpha) = \frac{1}{\gamma - 1}(\alpha^{\gamma-1} - 1) \quad (\gamma \neq 1) \tag{16}$$

We use a separate program to solve the above set of differential equations and generate the Bondi solution for density, pressure and velocity profiles, then use that to initialize our grid. The theoretical Bondi accretion rate is

$$\dot{M}_{\rm B} = 4\pi\lambda \frac{G^2 M^2 \rho_\infty}{c_{s,\infty}^3}\ . \tag{17}$$

For $\gamma = 5/3$, the parameter $\lambda = 0.25$.

We have conducted a number of runs with different accretion and reset radii, with the run parameters shown in Table 1. For convenience, P_∞ and ρ_∞ and M values we use resemble the Perseus cluster ($P_\infty = 3.2 \times 10^{-10}$ dyn cm^{-2}, $\rho_\infty = 6.58 \times 10^{-26}$ g cm^{-3}), and we use a fixed $\gamma = 5/3$. The gravitational potential is set to be a fixed point mass potential with a central SMBH of mass $M = 3.0 \times 10^9\ M_\odot$. With this setup, the maximum free fall timescale is $t_{\rm ff,max} \sim \sqrt{L^3/GM} \sim 1.11$ Myr.

Table 1. Table of runs conducted for the Bondi test. The last column describes what kind of resetting is done, where "Bondi" means resetting using the Bondi solution, and "Kernel" using our subgrid kernel method.

Run Number	$\Delta x/r_{\rm B}$	$R_{\rm cs}/r_{\rm B}$	$R_{\rm reset}/r_{\rm B}$	Method
W0	1/16	1	N/A	None
W1	1/16	1	1	Kernel
W1B	1/16	1	1	Bondi
W2	1/16	1	1/2	Kernel
W3	1/16	1/2	1/2	Kernel
P0	1/4	1	N/A	None
P1	1/4	1	1	Kernel
P1B	1/4	1	1	Bondi

Figure 2 depicts the flow for runs with different resolutions. In both well-resolved and poorly-resolved cases, without resetting the gas there is an accumulation of material around the center, eventually developing a back pressure that influences the flow of the gas outside the control surface and leads to an increase in the density and temperature there. However, by resetting the gas we are able to maintain a steady inflow, as is predicted by the theoretical Bondi solution. Moreover, while the kernel reset method knows nothing about the "correct" solution, it reduces the back pressure and converges to the Bondi-resetting result outside the control surface.

This can be demonstrated quantitatively using the radial gas profiles, measured from the center. In Figure 3 we show the deviation of these properties from the Bondi solution, since the difference in absolute values of various gas properties is very small. Through the profiles we can further see the influence of gas properties with and without resetting. With kernel resetting, although we see larger errors inside the control surface compared to resetting directly to the Bondi solution, for flow outside $R_{\rm cs}$ the difference is negligible (within 10^{-3} error). Higher resolution improves the agreement with the Bondi solution especially around the Bondi radius, most significantly in the flow velocity and pressure.

This result is further supported by the accretion rate, as shown in Figure 4. With resetting the accretion rate remains close to the Bondi accretion rate predicted by the analytical formula. The most accurate measurement of accretion rate comes from defining the control surface radius to be the same as the Bondi radius. If keeping the control surface fixed to be the Bondi radius, decreasing the resetting radius can decrease the error of the accretion rate calculation. In general, the best outcome comes from using the same value for the control surface